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Thermodynamics as a Science of Symmetry

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A new interpretation of thermodynamics is advanced; thermodynamics is the study of those properties of macroscopic matter that follow from the symmetry properties of physical laws, mediated through the statistics of large systems.

1. INTRODUCTION

The foundations of thermodynamics have been sought through many approaches: the empirical focus of the traditional Caratheodory formulation,[(1)] the axiomatic formulation, [(2)]the postulational approach,[(3,4)] the appeal to dynamical considerations,[(5)] and various others.

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Recent developments in many-body theory and field theory, in the theory of phase transitions, and in the general theory of symmetry provide a new perspective. It appears that symmetry considerations lie ubiquitously at the very roots of thermodynamic theory, so universally and so fundamentally that they suggest a new conceptual basis. In this paper we present an initial, yet-incomplete, contribution to the formulation of such a conceptual structure for thermodynamics.

The interpretation which this work suggests is that *thermodynamics is the study of those properties of macroscopic matter that follow from the symmetry properties of physical laws, mediated through the statistics of large systems.*

The fundamental laws of physics, some known and some perhaps remaining to be discovered, are specific and definite. But in addition to their detailed features they possess various symmetry properties. The existence of these symmetry properties alone, independent of the detailed content of the fundamental laws, imposes constraints and regularities on the possible properties of matter. These restrictions and regularities, together with the statistical features of large collections, are the content of thermodynamics.

In this perspective the generality, the model independence, and the peculiar nonmetric aspects of thermodynamic theory become natural and understandable.

Since all physical theories are logically tautological we may also expect the results of thermodynamic theory to reflect the character of its foundations. And so they do. Unlike other branches of physics, thermodynamics does not provide numerical predictions—it does not evaluate the specific heat or compressibility of any system. Instead it predicts that apparently unrelated quantities are equal $[(1/T)(dQ/dP)_T = -(dV/dT)_P]$, or that two coupled linear irreversible processes satisfy the Onsager reciprocity theorem ($L_{12} = L_{21}$). These are the types of results to be expected of a theory grounded in symmetry principles, and they retrospectively lend support to the symmetry interpretation of thermodynamics.

2. NOETHER'S THEOREM AND PHYSICAL SYMMETRY PRINCIPLES

We briefly recapitulate certain relevant aspects of physical symmetry principles, preparatory to a discussion of thermodynamics proper.

The primary theorem relating symmetry considerations to physical consequences is Noether's theorem.[2] According to this theorem, every

[2] See Hill.[6] For additional references see Ref. 7.

continuous symmetry transformation under which the Lagrangian of a given system remains invariant implies the existence of a conserved function (of the microcoordinates and their derivatives). That is, *every continuous symmetry of a system implies a conservation theorem, and vice versa.*[3]

The most primitive class of symmetries[9] is the class of *continuous space–time transformations.* The (presumed) invariance of physical laws under time translation implies the conservation of energy. Symmetry under spatial translation implies conservation of momentum, and rotational symmetry implies conservation of angular momentum.

The discrete space–time symmetries (of time reversal, spatial inversion, and charge conjugation) appear to be rigorous symmetries of the strong and electromagnetic interactions, although they are violated by the "weak interaction." Accordingly, they are applicable to common thermodynamic systems, in which weak interaction processes (e.g., beta decay) play no significant role. Thermodynamic theory would require appropriate modifications for astrophysical systems in which weak interaction processes might be important.[10]

An important class of symmetries has been termed "dynamical."[9] The symmetry of the electromagnetic equations under gauge transformation is the prototype of the class of dynamical symmetries. It refers to a particular mathematical transformation of the electromagnetic equations (acting on the scalar and vector potentials and on the operators of the matter field) which leaves the equations invariant. This abstract mathematical quality is characteristic of dynamical symmetries; they are "formulated in terms of the laws of nature," in contrast to the geometrical symmetries, which can be "formulated in terms of the events themselves."[9] Thus the geometrical symmetry principle of spatial translation can be stated in terms of the equivalence of two processes occurring at different places, but no such physical interpretation can be given to the changes in the scalar and vector potentials that leave Maxwell's equations invariant.

The conserved quantity corresponding to gauge symmetry is the electric charge.

Other dynamic symmetries give rise to the conservation of baryon number and of lepton number.

Many additional symmetries exist, such as the symmetries giving rise to the conservation of strangeness and isospin in strong interactions. Because the associated nuclear processes generally do not occur in conventional thermodynamic systems, these symmetries do not play a role in current thermodynamic theory.

[3] Limitations on the theorem, particularly on its inverse, are discussed by Rosen.[8]

3. BROKEN SYMMETRY AND GOLDSTONE'S THEOREM

Another important area of symmetry considerations centers on the concept of *broken symmetry*.[11–13] We summarize the relevant aspects, following the review of Grib *et al.*[11]

A system *may* exhibit the full symmetry of its Lagrangian, but it is characteristic of infinitely large systems that they also may "condense" into states of lower symmetry. As has been stressed also by Anderson,[13] this leads to an essential difference between infinite systems and finite systems, such that the former do not satisfy the laws of quantum mechanics in the simple manner of finite systems.[11]

The broken symmetry state is distinguished by the appearance of a "macroscopic order parameter." This macroscopic order parameter obeys the laws of classical physics, as distinct from the laws of quantum mechanics.

Whereas all quantum states of a finite system lie within a single Hilbert space, the appearance of a macroscopic order parameter determines a continuum of disjoint Hilbert spaces. To each macroscopically distinct value of the order parameter there is a separate Hilbert space. All the states in one such Hilbert space are orthogonal to all the states in another. The transformation from one Hilbert space to another is not unitary, and in each space the microobservables are described by operators which realize unitarily nonequivalent representations of the observables. These facts manifest themselves physically in the absence of interference between states with different values of the macroscopic order parameter.

The breaking of the symmetry establishes a multiplicity of "vacuums" or ground states, related by the transformations of the (broken) symmetry group. The various values of the macroscopic order parameter are in a many-to-one correspondence with the several ground states (e.g., each value of the vector magnetic moment of a spherical ferromagnet is associated with that ground state with the same direction of its magnetic moment). According to Coleman's theorem, this ground state exhibits the maximal lowering of the symmetry of all its associated macrostates, or "everything broken for the world is broken for the vacuum.[11]

Finally, broken symmetry systems are subject to Goldstone's theorem,[11,14] which states that if the broken symmetry group is a continuous group whose generators can be written as integrals of local densities, and if the interaction falls off sufficiently rapidly with distance, then there exists a branch of elementary excitations for which the energy goes to zero in the long-wavelength limit.

4. THERMODYNAMIC COORDINATES; CONSERVED COORDINATES

The scope of thermodynamics is determined by the criteria for selection of the variables of the theory. The criteria are suggested, but not demanded, by the nature of conventional macroscopic measurements.

Macroscopic measurements generally are exceedingly slow with respect to the atomic scale of time (typically of the order of 10^{-15} sec) and exceedingly coarse with respect to the atomic scale of distance ($\sim 10^{-8}$ cm). Furthermore, macroscopic measuring devices usually yield spatially integrated responses; they respond additively to the subregions within their region of sensitivity. Accordingly, *thermodynamic coordinates are defined as spatially homogeneous, extensive* (*additive*), *and time independent*. This definition severely limits the states to which thermodynamics can apply and it excludes all dynamical considerations.

The most obvious candidates for thermodynamic coordinates are those extensive quantities that are *conserved*, this being the most obvious way (but *not* the only way) of achieving time independence. Each such conserved coordinate bespeaks an underlying physical symmetry.

The most immediately evident conserved coordinate is, of course, the *energy* (time-translation symmetry). Its relevance as a thermodynamic coordinate underlies the "first law" of thermodynamics.

Time-translation, spatial translation, and spatial rotation symmetries are interrelated in a single class of continuous space–time symmetries. The symmetry interpretation of thermodynamics immediately suggests, then, that energy, linear momentum, and angular momentum should play fully analogous roles in thermodynamics. The equivalence of these roles is rarely evident in conventional treatments, which appear to grant the energy a misleadingly unique status. The momentum and the angular momentum are generally suppressed by restricting the theory to systems at rest, constrained by external "clamps." Nevertheless, it is evident that in principle the linear momentum does appear in the formalism in a form fully equivalent to the energy, for relativistic considerations[4] imply that the energy in one frame appears partially as linear momentum in another frame. Similarly, the angular momentum is only occasionally introduced explicitly into thermodynamic formalisms (as in astrophysical applications to rotating galaxies); it appears, for instance, in the "Boltzmann factor," $\exp(-\beta E - \beta \mathbf{\Lambda} \cdot \mathbf{L})$, additively and symmetrically with the energy.

To stress these facts we might well amend the first law to read that "the extended first law of thermodynamics is the symmetry of the laws of

[4] For a recent review see Ref. 15. Also see Ref. 16.

physics under space and time translations and under spatial rotation."

The equality of status of the energy, the momentum, and the angular momentum dispels the separate and unique role generally accorded to the energy. Thermodynamics is not appropriately characterized as "the science of energy and of its transformations." To dramatize this fact, we exhibit, in the appendix, a simple, specific model for which a full thermodynamic theory can be formulated. In this model the energy is totally absent as a relevant variable. Familiar central formulas such as $dS = dQ/\tau$ remain in the theory, although dQ does not have the significance of heat or energy.

5. OTHER CONSERVED COORDINATES

We have identified conserved quantities as forming one class of physical variables satisfying the criteria for thermodynamic coordinates. More specifically, we have identified that particular set of conserved variables that originate in the continuous space–time symmetries of physical laws. Other symmetry principles result in additional conserved parameters.

In a system of particles undergoing the full range of possible nuclear interactions the *composition* of the system would be described by the conserved baryon and lepton numbers. These are the fundamental compositional coordinates of thermodynamics. In practice, however, much stronger approximate conservation theorems apply. The number of atoms of each atomic species, or the number of molecules of each molecular species, may be approximately conserved. In such cases the *mole numbers* are appropriate compositional coordinates.

Even where the conservation rules underlying the compositional coordinates are only approximate, they are generally grounded in symmetry considerations.[17] An instructive example is the case of ortho- and para-hydrogen. Para-hydrogen molecules have antiparallel nuclear spins and ortho-hydrogen molecules have parallel nuclear spins; they are antisymmetric and symmetric, respectively, under permutation of nuclei. This difference in symmetry gives rise to a *selection rule*; the two molecular species thereby are individually conserved, and the proper thermodynamic coordinates include the mole numbers of ortho-hydrogen and of para-hydrogen *separately*. Only by including both mole numbers as independent variables can agreement between thermodynamic theory and experiment be attained. However. addition of a small admixture of water vapor or of oxygen alters the situation drastically. The magnetic field of the paramagnetic oxygen ions distorts the symmetry of the hydrogen ions and thereby destroys the selection rule. Ortho- and para-hydrogen then freely intraconvert, and a proper thermodynamic description includes only a single compositional variable—the

mole number of hydrogen. Thus the proper choice of the thermodynamic coordinates describing the composition of a thermodynamic system is based on conservation theorems reflecting underlying symmetry considerations.

Under appropriate conditions all of the conservation theorems of physics would be reflected in associated thermodynamic coordinates. The total charge and the total "strangeness," for instance, would join the energy, angular momentum, hyperon numbers, etc., as independent variables in the thermodynamic fundamental equation.

6. BROKEN SYMMETRY COORDINATES

We continue the survey of variables which qualify as thermodynamic coordinates (knowing full well that we must eventually arrive at the familiar list $U, V, \mathscr{M}, N_1, N_2, \ldots$). The energy U is the prototype of variables conserved by virtue of the continuous space—time symmetries; the mole numbers N_1, $N_2, \ldots$ are the prototypes of variables conserved by virtue of various other symmetries. In addition to these there are yet additional variables which, though not conserved, nevertheless are time independent. These are the broken symmetry coordinates discussed in Section 3. The prototypes of these variables are the electric dipole moment $\mathscr{P}$, or the magnetic dipole moment $\mathscr{M}$, or the volume V.

Broken symmetry coordinates are associated with the order parameters of condensations from states with the full symmetry of the Lagrangian to states with a lesser symmetry. They are classical parameters, obeying classical mechanics, and determining disjoint Hilbert subspaces of the full Hilbert space of the system.[11]

The macroscopic role of the broken symmetry coordinates has been explored most fully by Halperin and Hohenberg[18] and Martin *et al.*[19] These authors first observe that the macroscopic dynamics of systems is describable in terms of certain long-lived degrees of freedom or modes. Martin *et al.* state[19] that "the existence of such modes can always be traced either to conservation laws or, in the case of ordered systems, to 'continuous broken symmetries.' The macroscopic dynamics then can be characterized by hydrodynamic considerations," which Halperin and Hohenberg[18] define as theories "utilizing only the conservation laws for the Hamiltonian, the symmetry properties of the ground state, and assumptions that one can expand certain quantities in powers of the gradients and magnitudes of the deviations from equilibrium". The thermodynamic theories in which we are interested are the static limits of these hydrodynamic theories.

Consider specifically the magnetic moment of a ferromagnet. The Hamiltonian of an isolated system is necessarily rotationally symmetric, but

at a particular "Curie temperature" the elementary magnetic moments of a system may "condense" into a state of broken symmetry. The system then develops a *macroscopic* magnetic moment, in some arbitrary direction (taken as the z axis). Assume the magnitude of the magnetic moment to be $\mathscr{M} = mM_0$, where M_0 is the saturation value of the magnetization. Rotation of any *finite* number of spins does not alter the value of m (in the $N \to \infty$ limit) but the corresponding states define a Hilbert space. All states in this space are orthogonal to all states corresponding to some other value of m, which differ by rotation of an *infinite* number of spins.[11] In addition, all states excited from the ground state along z are orthogonal to all states excited from the ground state oriented at some angle θ, for the scalar product of the two states is proportional to $(\cos\theta)^N$, which vanishes as $N \to \infty$. Both the direction and magnitude of the magnetization are classical variables by virtue of the properties conferred on them by the principles of broken symmetry.

We shall see later that the Goldstone spectrum of excitations underlies the unique properties of the phase transitions at which broken symmetry coordinates make their appearance.

The multiplicity of ground states of a ferromagnet (oriented in various directions) is another common feature of broken symmetry coordinates. *This ground-state degeneracy does not contribute to the entropy at zero temperature, however, because of the noninterference of states with distinct order parameters.*

Thus a variety of important thermodynamic properties are linked to the broken symmetry origin of thermodynamic coordinates such as the magnetic moment.

It should be mentioned in passing that in a common model, consisting of an array of interacting electron spins, the magnetic moment is *conserved* by virtue of angular momentum conservation. This accidental and misleading property never applies to a real system. If the system contains both electron spins and orbital moments, each with a different g-factor, or if the electron spins interact with the lattice through spin–orbit coupling, the magnetic moment is *not* a conserved coordinate.

The electric dipole moment is another broken symmetry coordinate, closely analogous to the magnetic moment.

The number of particles (or mole numbers) in the condensed phase of a superfluid,[12] the sublattice magnetization of an antiferromagnet,[12] and the order parameter of a binary alloy are additional broken symmetry coordinates.

Finally, the *volume* must be understood as a broken symmetry coordinate. A finite volume of a solid or liquid system breaks the continuous translational symmetry of space. A gas has no intrinsic value of the volume; again, the

translational symmetry of space must be broken by the solid system which contains the gas. Perhaps more fundamentally, any finite density of particles defines a unitarily disjoint Hilbert space, and it breaks the symmetry under Galilean transformations.[11]

The broken symmetry origin of the volume is evident in the characteristic properties of the condensation of a solid or liquid system. Such a second-order transition is fully analogous to the ferromagnetic transition, exhibiting similar critical behavior. The Goldstone excitations are the longitudinal phonons (analogous to the spin waves of a ferromagnet), and it is the infinite fluctuations of these excitations which dominate the transition.

The role of the volume as a broken symmetry coordinate is revealed even more clearly in the thermodynamics of anisotropic crystals. There the components of the (finite) strain tensor (relative to some arbitrary reference system) play the role of thermodynamic coordinates. They define the anisotropic shape of the system, and they clearly are broken symmetry coordinates. In simple fluid systems the off-diagonal elements necessarily vanish and the diagonal elements are equal; only the trace of the strain tensor remains as an independent variable. That trace is the volume.

It is a convenient happenstance that the conventional "simple systems" discussed in elementary thermodynamics have one prototype of each of the three characteristic classes of thermodynamic coordinates; coordinates conserved by the continuous space–time symmetries (U), coordinates conserved by other symmetry principles (N), and nonconserved, broken symmetry coordinates (V).

7. THE SEPARATE ROLES OF CONSERVED AND BROKEN SYMMETRY COORDINATES

In this synoptic survey of thermodynamics we implicitly adopt the logical framework of the postulational approach.[4] That is, the extensive coordinates U, V, N,... are taken as fundamental, in agreement with our understanding that such coordinates are mechanical rather than statistical; they can be defined in principle for every microstate, or for nonequilibrium macrostates, and they can be represented by quantum mechanical operators. Of course these reasons lie outside macroscopic phenomenology, but the logical exercise of erecting a theory without external references is not our present purpose.

The postulatory formulation then confronts the basic typical problem of thermodynamics. Suppose two isolated systems with parameters U_1^0, V_1^0, N_1^0,... and U_2^0, V_2^0, N_2^0,... are brought into diathermal and semipermeable contact; that is, the energy U and the partial mole number N are permitted to

redistribute between the two systems, whereas the volumes, the mole numbers of other components, and all other coordinates are maintained separately constant. *It is postulated that the solution of such problems has the form characteristic of most areas of theoretical physics—namely that the solution is provided by an extremum principle.*

To recapitulate the consequences briefly, it is postulated that there exists an extensive function of the extensive coordinates (called the entropy) which is maximum in the equilibrium state. Then to solve the problem above we write (by the extensivity property of the entropy) that

$$S = S_1(U_1, V_1^0, N_1^0, \ldots) + S_2(U_2, V_2^0, N_2^0, \ldots) \tag{1}$$

By the extremum principle

$$\delta S = \frac{1}{T_1}\,\delta U_1 - \frac{\mu_1}{T_1}\,\delta N_1 + \frac{1}{T_2}\,\delta U_2 - \frac{\mu_2}{T_2}\,\delta N_2 = 0 \tag{2}$$

where, *by definition,*

$$1/T \equiv \partial S/\partial U; \qquad -\mu/T \equiv \partial S/\partial N \tag{3}$$

Conservation of energy and of mole number implies

$$\delta U_1 + \delta U_2 = 0; \qquad \delta N_1 + \delta N_2 = 0 \tag{4}$$

or

$$\delta S = \left(\frac{1}{T_1} - \frac{1}{T_2}\right)\delta U_1 - \left(\frac{\mu_1}{T_1} - \frac{\mu_2}{T_2}\right)\delta N_1 = 0 \tag{5}$$

It follows that the temperatures of the two systems are equal ($T_1 = T_2$) and that the chemical potentials of the permeable component are equal ($\mu_1 = \mu_2$). The equations of state (3) determine the quantities T_1, T_2, μ_1, and μ_2 as functions of the extensive parameters, permitting evaluation of the values of these extensive parameters in the equilibrium state. A by-product of the formalism is the identification of the quantity T (introduced formally as a partial derivative of the fundamental equation) with the intuitive notion of the temperature.

Now this particular method of introducing the entropy and the intensive parameters is only one of many alternative approaches, most of which proceed inversely, in the much more difficult direction from processes to state functions. The postulatory framework is *not* essential to the symmetry interpretation which is our present interest. Nevertheless its unique simplicity makes it particularly convenient for a discussion of the nature of thermodynamics, and specifically at this moment for the roles of different classes of thermodynamic coordinates.

In the problem above each of the two free coordinates (energy and mole number) is a conserved rather than a broken symmetry coordinate. In contrast, consider two systems with magnetic moments $\mathscr{M}_1$ and $\mathscr{M}_2$. A problem analogous to that above does not make physical sense. We do not normally have a problem of "interchange" of magnetic moment between two interacting systems, and certainly not subject to a conservation theorem on the total magnetic moment. The analog of the problem above does not arise for the electric dipole moment, for the order parameter of superfluid helium, or for any other broken symmetry coordinate.

Now the volume might at first seem to be an exception to these statements. A standard problem of thermodynamics involves two gases separated by a movable piston, contained within a common cylinder. Motion of the piston then leads to $\delta V_1 = -\delta V_2$, and an analysis identical to Eqs. (1)–(5) leads to equality of temperature and pressure. The "conservation" equation ($V_1 + V_2 = \text{const}$) here is *not* of a fundamental nature, however. Suppose that gas 1 is in a cylinder of cross section A_1, closed by a movable piston; gas 2 is in a *different* cylinder of cross section A_2, also fitted with its own piston. We connect the two pistons by a rigid bar. Then the "conservation" relation is replaced by

$$\delta V_1/A_1 = -\delta V_2/A_2 \tag{6}$$

The solution leads to the self-evident result $P_1A_1 = P_2A_2$.

Hence we see that broken symmetry coordinates are subject to very different auxiliary conditions than are conserved coordinates. In some cases, as in Eq. (6), we are free to devise various constraints on a broken symmetry coordinate. In the case of the magnetization of a ferromagnet we can control the intensive parameter (the magnetic field); the auxiliary condition is then that the partial derivative $(\partial S/\partial \mathscr{M})_{U,V,N}$ must have a specified value. In the case of an antiferromagnet the intensive parameter (a staggered field) is not subject to control, and the auxiliary condition is that the derivative $\partial S/\partial \mathscr{M}_{\text{subl}} = 0$. Similarly the intensive parameter corresponding to the quantity of superfluid component in liquid helium has no physical realization and must be set equal to zero. *In each case a broken symmetry coordinate is subject to an external auxiliary condition, in contrast to conserved coordinates which are determined by universal conservation conditions.*

8. THE ENTROPY-MAXIMUM PRINCIPLE AND "UNITARITY SYMMETRY"

We have stressed, with more than usual emphasis, the physics underlying the initial choice of variables of thermodynamic theory. That choice is a

fateful one. A very few variables meet the criteria of extensivity and time independence required for macroscopic measurability. Those which qualify do so by virtue of one or another symmetry consideration. All other microscopic variables thereby are consigned to a class of incoherent, nonobservable modes which underlie the concept of heat. Analysis of the symmetry origin of the thermodynamic coordinates has led us also to the first law of thermodynamics. It remains to probe more deeply into the second and third laws, to examine the emerging theory of second-order phase transitions, and to explore the origin of the Onsager extension to irreversible processes.

We shall examine these seminal princples of thermodynamics in turn, to reveal their symmetry origins. Integration of the entire structure into a coherent and complete exposition of thermodynamics is not undertaken in this paper, since it would require repetition of an elaborate formalism with which the reader presumably is familiar.

The thermodynamic form of the entropy principle, as distinct from the statistical mechanical form, is based on the *equal a priori probability of states*. The thermodynamic principle is purely static. It is rooted in the principle that *in equilibrium* all permissible microstates contribute equally to the macrostate, and that the entropy is simply $k_B \log W$, where W is the number of permissible microstates.

Mechanisms of irreversibility, or of entropy increase, are *dynamical*, and they lie outside the scope of thermodynamics. Nevertheless the entropy-maximum property of equilibrium states is hardly understandable unless linked with dynamical considerations, and we shall comment on these in Section 10. For the moment we restrict our attention strictly to static or thermodynamic considerations.

The equal a priori probability of states is already in the form of a symmetry principle, even before we trace its origin to deeper principles. The entropy depends *symmetrically* on all permissible states. This principle, if accepted, determines the functional form of the entropy; the only symmetric function of a set (in this case a set of states) is the number in the set, or a function of the number. The particular function $S = \log W$ is determined completely then by symmetry over the set of states and by the requirement of extensivity.

However, we still seek the origin of the equal a priori probability of states in a deeper symmetry. It would appear that the fundamental symmetry is a *dynamical* symmetry;[9] specifically the invariance of the equations of statistical mechanics under a rotation in the physical Hilbert space (that is, under arbitrary unitary transformations among the physically permissible microstates). *The conserved quantity resulting from this symmetry is the volume in phase space* W (or the log of W).

The conservation of the ("fine grained") phase space volume under an

internal unitary transformation is, of course, the famous Boltzmann H-theorem. That theorem is usually stated in terms of one particular unitary transformation—the time translation operation—but it depends only on the unitarity of the transformation.

It is interesting to reflect on the significance of the conservation of W. An arbitrary ("fine grained") phase space distribution propagates with time, flowing through phase space without either coalescing or diffusing. The probability density tends neither to coalesce around particular favored states in phase space, nor does it diffuse to avoid particular unfavored states. *There are no "favored" states in quantum mechanics, and no "disfavored" ones, and this impartiality is a dynamical symmetry of the quantum mechanical formalism.*

Unitarity is often characterized simply as "conservation of probability," but it is considerably more. A transition matrix of the form

$$\begin{pmatrix} 1/2 & 1/2 \\ 0 & 1 \end{pmatrix}$$

conserves probability; a system in state 1 either remains in state 1 or makes a transition to state 2, each with probability $\frac{1}{2}$; a system in state 2 necessarily remains in that state. Probability is conserved, but the second state is favored over the first, and an arbitrary probability density coalesces entirely into the second state. The unitarity symmetry of quantum mechanics (or of classical mechanics) disbars such asymmetric transition matrices.

Given unitarity symmetry, or the impartiality of quantum mechanics among the microstates, it follows that a uniform probability density in phase space remains uniform under the intrinsic dynamics of the system.

Furthermore, a uniform distribution is the only distribution which is both stationary and *stable with respect to random, external perturbations.*

It follows that a system in equilibrium, stationary in time, and stable under random fluctuations of its environmental interactions is represented by a distribution function which is uniform in the permissible region of phase space. This is the crux of the thermodynamic entropy principle.

9. TIME-REVERSAL SYMMETRY

It may contribute to clarity to restate some of the foregoing observations in the illuminating (but nonrigorous) formalism of the Pauli master equation,[20,21]

$$\frac{d\rho_i}{dt} = \sum_j (\omega_{ij}\rho_j - \omega_{ji}\rho_i) \tag{7}$$

Here ρ_i is the probability of occupation of state i (ρ_i is the diagonal element of the density matrix) and ω_{ij} is the transition probability from state j to state i. (All off-diagonal elements of ρ are neglected.) The unitarity condition on the S-matrix (the ω_{ij} are squares of the S-matrix elements) requires

$$\sum_j \omega_{ij} = \sum_j \omega_{ji} \tag{8}$$

It follows immediately that a solution of Eq. (8) is the uniform probability distribution

$$\rho_i = \text{const} \tag{9}$$

As has often been pointed out, time-reversal symmetry, or the condition

$$\omega_{ij} = \omega_{ji} \tag{10}$$

is a stronger condition than unitarity and, it leads "independently" to the same solution. We shall not enter into a discussion of the conditions for time-reversal symmetry.[23] But we stress that *the interchangeability of the unitarity condition* (8) *and time-reversal symmetry* (10) (when the latter is applicable) *demonstrates that unitarity is properly viewed as a symmetry condition.* Either of these symmetry conditions underlies the equal a priori probability of states (9), and thence the entropy-maximum principle of thermodynamics.

10. DYNAMICAL ASPECTS OF THE SECOND LAW

As stressed previously, dynamical aspects lie outside the domain of thermodynamics, but a discussion of the mechanism of irreversibility nevertheless clarifies the static concepts.

In our discussion of the logical foundations of the static entropy principle we pointed out that various delicately contrived distributions may be time independent, but that they generally are unstable with respect to random external perturbations. The uniform distribution is selected because of its stability to such perturbations. This argument does *not* necessarily imply that these external perturbations are the *physical* mechanism leading to the uniform distribution. Random perturbations provide a *sufficient* logical basis for the conclusion, but they are not necessarily the *operative* mechanisms.

It would appear that three separate aspects of the dynamical approach to equilibrium should be clearly distinguished. These aspects are (a) *the approach to equilibrium properties*, (b) "*effective irreversibility*," and (c) "*absolute irreversibility*."

The approach to equilibrium properties appears to be a consequence of the *local* property of observable quantities, which generally depend only on one-particle and two-particle operators. Long before a system traverses its complete phase space these local operators apparently tend to assume their terminal values. Swendsen[24] has carried out a calculation of the one-dimensional, noninteracting gas for which the "container" is suddenly expanded to twice its original length. In approximately one transit time the density becomes uniform (in 1 mm regions) to an accuracy of 10^{-6} (for $L \simeq 1$ m and $N \simeq 10^{15}$). In this case there is a rapid approach to equilibrium properties under conditions (no interparticle collisions) precluding irreversible behavior, and in a time short compared to the time required for the system to pass through all its permitted microstates. A similar insight is provided by computer calculations of the time evolution of interacting gases; these show that local hydrodynamic flow patterns (velocity–velocity correlation functions) are well established after only approximately ten interparticle collisions.[25]

A slower process is the development of *effective irreversibility*. Sinai's famous theorem[26] on the mixing property of a gas of even two particles has shown that the time evolution of the probability density in phase space keeps its total volume constant, but ramifies it out into a maze of fine strands which are quasidense in phase space. In principle such an evolution could be reversed, but in practice it is outlandish to visualize the arrangements required to prepare such a reverse reaction. An example is provided by the spontaneous $2p \rightarrow 1s$ transition in a hydrogen atom, with the emission of a photon. The photon is radiated in all directions in space, with relative phases depending on the initial direction of the angular momentum in the $2p$ electronic state. This simple atomic process is reversible *in principle*, as are all fundamental quantum mechanical processes. But how is it *operationally* reversible? Simply sending in a photon from some particular direction to be absorbed in the $1s \rightarrow 2p$ transition is *not* sufficient, for it will result at best in a $2p$ state with some accidental direction of orbital angular momentum, unrelated to the initial direction. We must prepare an incoming spherical wave with the proper phase relations over the entire wavefront. This is so utterly impractical that for all reasonable purposes the initial radiation can be thought of as *operationally irreversible*. How much more so is any macroscopic process! True reversal generally would require such outlandishly complex preparations as to make a travesty of the claim of "reversibility in principle."

Effective irreversibility is a consequence of the imposition of highly specialized initial conditions, and of the complexity of macroscopic systems.

After the "mixing property," or the ramification of the probability density quasidensely through phase space, a third process necessarily occurs. This is the destruction of the coherence of the system by external perturba-

tions. These perturbations may be real incursions, such as cosmic rays, or virtual interactions with distant systems through vacuum fluctuations. A gas in a laboratory in Columbia University would be "irreversibly perturbed" in 1 msec by a person shifting in his chair in Chicago (an irreversible perturbation here being estimated as transfer of an impulse sufficient to induce a transition between two single-particle states)! More fundamentally, these random perturbations may be connected with cosmic processes and the expansion of the universe.[27] They transform the *effective irreversibility* described in the previous paragraph into true or *absolute irreversibility* (probably without any discernable effects on the system).

11. PHASE TRANSITIONS AND BROKEN SYMMETRY

Much of the conventional wisdom of thermodynamics has been shaken by recent developments in the theory of phase transitions.[28] The unexpected singularities of thermodynamic functions at critical points and the universality of the critical point exponents challenge the scope of classical theory.

Second-order transitions are associated with the breaking of the symmetry of the Lagrangian of the system, and with the appearance of broken symmetry coordinates. Phase transitions therefore are a central and fundamental aspect of the thermodynamic formalism, directly coupled to the very existence of the basic thermodynamic coordinates.

For simplicity of discussion we consider a second-order transition in a crystal, thinking, for instance, of the onset of a ferroelectric moment. Above the transition the charge density in the crystal is invariant under some symmetry group $\mathscr{G}$. In the classical theory of Landau[29],[5] one expands the charge density in a set of basis functions of the irreducible representations of the group $\mathscr{G}$:

$$\rho(\mathbf{r}) = \rho_0(\mathbf{r}) + \sum_{\Gamma,m}{}' \rho_{\Gamma,m}\phi_{\Gamma,m}(\mathbf{r}) \tag{11}$$

Here $\rho_0(\mathbf{r})$ belongs to the fully symmetric representation, the sum is over all *other* representations Γ and over the row index m of each representation. The $\phi_{\Gamma,m}(\mathbf{r})$ are the basis functions. The free energy can then be considered as a function of the coefficients $\rho_{\Gamma,m}$. The form of the function is dictated by the symmetry $\mathscr{G}$; it must be composed of invariant combinations of the $\rho_{\Gamma,m}$, whence

$$F = F_0 + \sum_{\Gamma,m}{}' A_\Gamma(T, P,\ldots)\rho^2_{\Gamma,m} + \text{4th-order invariants} + \cdots \tag{12}$$

[5] Also see Ref. 30, Chapter 14.

At high temperature the stability of the symmetric state implies that all the A_Γ coefficients must be positive. Landau postulated that, as the temperature is lowered to some "critical" value T_c, some particular one of the coefficients A_Γ passes through zero. Below the temperature T_c the coefficient $A_\Gamma(T, P; N)$ is negative. Then minimizing the free energy (12) leads to nonzero values of the quantities $\rho_{\Gamma m}$, the actual values depending on the higher-order terms in Eq. (12). It is well known that this theory leads to singularities of a form which disagrees with experiment.[31]

Michelson[32] has emphasized that the symmetry argument above is fatefully oversimplified in that it assumes that a single coefficient A_Γ vanishes alone. This is, in fact, forbidden by the continuity within the symmetry group of a real system. That symmetry group is a *space group*, containing both translational and rotational (or reflection) elements.[29] The translational subgroup is Abelian, with one-dimensional representations labeled by the wave vector **k**. Then Eq. (12) becomes

$$F = F_0 + \sum_{\mathbf{k}} A_{\mathbf{k}}(T, P,\ldots)\rho_{\mathbf{k}}{}^2 + \sum_{(\mathbf{k}_1\mathbf{k}_2\mathbf{k}_3\mathbf{k}_4)} B_{1234}(T, P,\ldots)\rho_{\mathbf{k}_1}\rho_{\mathbf{k}_2}\rho_{\mathbf{k}_3}\rho_{\mathbf{k}_4} + \cdots \tag{13}$$

Restrictions imposed by the space group symmetries imply that $A_{\mathbf{k}}$ and B_{1234} are symmetric under various rotations or reflections in k space. Assume that the instability occurs at $k = 0$, which is the case of most general interest.[29] Then the coefficient A_0 passes through zero, and the continuity of the $A_{\mathbf{k}}$ ensures that neighboring coefficients have neighboring zeros. As the temperature is lowered additional $A_{\mathbf{k}}$ continually become unstable, and the singularity is formed by the confluence of a sequence of singular points. Expanding the coefficients around zero wave vector ($A_k = A_0 + A'k^2 + \cdots$) and recasting Eq. (13) in real space, we find

$$F = F_0 + \int d^3r\, \{A'[\Delta\rho(\mathbf{r})]^2 + A_0\rho^2(\mathbf{r}) + B\rho^4(\mathbf{r})\} \tag{14}$$

This, of course, is the Ginzburg–Landau free energy functional[34] on which the Wilson theory[26] is based. The essential singularity precludes the steepest-descent evaluation of the partition function (or the minimization of the free energy function). It will be recalled, however, that the more fundamental canonical average which is required is the result of an application to open subsystems of the basic principle of equal a priori probabilities of states.[35]

The intrusion of the various "modes" $\rho(\mathbf{k})$ into thermodynamic theory, demanded by the symmetry analysis, is rooted in even-deeper symmetry origins. These modes are, of course, the Goldstone excitations[11–14] associated with the broken symmetry coordinate emerging at the transition.

The consequences of the canonical averaging are realized in a specific functional structure of the free energy in the vicinity of the transition point. The divergent part of the free energy is shown to be a generalized homogeneous function of its variables (the "scaling form"). From this functional form all the scaling relationships and the critical exponents follow.[28,31]

Even the specific evaluation of the canonical average is not essential to all the results, and it is of interest to note that many of the results can be obtained by group-theoretical methods, using the concepts of the "renormalization group."[28]

12. ONSAGER RECIPROCITY

Except for the "third law," the seminal principles of equilibrium thermodynamics now have been discussed. Each derives from underlying symmetry considerations. The third law remains somewhat of a mystery. In its most naive form, expressed in terms of ground-state degeneracy, it is suggestive of a symmetry statement, since degeneracy in quantum mechanics is always associated with symmetry.[37] But a more careful analysis reveals[38,39] that the third law depends on the form of the density-of-states function at nonzero energy. The possible qualitative bases have been discussed elsewhere,[39] but the "law" remains an unexplained postulate.

In 1931 Onsager[40] extended thermodynamic theory beyond the realm of equilibrium, to include steady-state, linear processes. It is too well known to require elaboration here that Onsager based that extension on a single additional principle: the time-reversal symmetry of physical laws. He associated macroscopic processes with the decay of spontaneous fluctuations in an equilibrium system, and he assumed equality of the two time-reversed correlation functions:

$$\langle x_1(0)x_2(t)\rangle = \langle x_1(0)x_2(-t)\rangle \tag{15}$$

From this he was able to show the well-known reciprocity relationship between the mutual interference of two linear processes. The association of linear irreversible response with spontaneous fluctuations was later validated formally through the fluctuation-dissipation theorem,[41–43] and independent derivations of the Onsager reciprocity were given in terms of quantum statistical perturbation theory. In all of these derivations it was shown that the central basis of the Onsager theory is the time-reversal symmetry of physical laws.

13. CONCLUSION

In this sketchy outline we have probed the various germinal laws of thermodynamics. Two underlying principles consistently emerge. The bases of macroscopic thermodynamic laws lie in the symmetries of microscopic physical laws. And the irrelevant details of specific systems are blurred by the properties of large statistical collections (in analogy with the obliteration of detail under the central limit theorem). We conclude that thermodynamics properly can be defined as the study of those properties of macroscopic matter that follow from the symmetry properties of physical laws, mediated through the statistics of large systems.

APPENDIX

To emphasize the lack of uniqueness of the energy in thermodynamics, we demonstrate a model in which the energy plays no relevant role. The model is, of course, "contrived," but its implications should not thereby be compromised.

Consider a system with N noninteracting spins, each of which can be "up" or "down," and each carrying one unit of angular momentum (in units of $\hbar/2$). The system has a moment of inertia I and an angular frequency of net rotation ω, so that the total angular momentum (z component) is

$$L = L_m + n_\uparrow - n_\downarrow \tag{A.1}$$

where

$$L_m = I\omega \tag{A.2}$$

and

$$n_\uparrow + n_\downarrow = N \tag{A.3}$$

Assume the moment of inertia to be infinitely large and ω to be infinitely small, so that $I\omega$ is finite. Then the kinetic energy of rotation $\frac{1}{2}I\omega^2$ vanishes. Furthermore the electron spins are noninteracting. Hence *all states of the system have zero energy, and the energy is not a relevant variable.*

The total angular momentum L is conserved and it can be measured by exploiting that conservation, precisely as the energy is usually measured in thermodynamics.[4] Similarly, the mechanical angular momentum is observable (both I and ω are directly measurable). We disbar magnetic effects (as if the electrons were uncharged, or magnetic fields screened), and the spin angular momentum $n_\uparrow - n_\downarrow$ is assumed to be *unobservable.*

The thermodynamic fundamental equation is of the form

$$S = S(L, L_m, N) \tag{A.4}$$

and the intensive parameters are defined by

$$\frac{1}{\tau} = \frac{\partial S}{\partial L}, \qquad -\frac{P_m}{\tau} = \frac{\partial S}{\partial L_m}, \qquad -\frac{\mu}{\tau} = \frac{\partial S}{\partial N} \tag{A.5}$$

with

$$\delta S = \frac{1}{\tau}\,\delta L - \frac{P_m}{\tau}\,\delta L_m - \frac{\mu}{\tau}\,\delta N \tag{A.6}$$

We can calculate the fundamental equation easily, as

$$S = k_B \ln \frac{N!}{n_\uparrow!\, n_\downarrow!} = k_B \ln \frac{N!}{[(N + L - L_m)/2]!\,[(N - L + L_m)/2]!} \tag{A.7}$$

$$\simeq -Nk_B \ln\left[1 - \left(\frac{L - L_m}{N}\right)^2\right] - \frac{1}{2}\,k_B(L - L_m) \ln \frac{N + L - L_m}{N - L + L_m} \tag{A.8}$$

The various equations of state can be calculated directly, giving $\tau = \tau(L, L_m, N)$, etc. We find that $P_m = 1$, this trivial *equation of state* resulting from the simplicity of the model; specifically that the incoherent modes of the spins are independent of the coherent mode of mass rotation (rather than coupled together as the incoherent phonon modes are coupled to the volume mode in a real crystal). Inserting this equation of state, we find from Eq. (A.6) that

$$\delta S = \delta Q/\tau \qquad \text{if} \quad \delta N = 0 \tag{A.9}$$

where

$$\delta Q = \delta L - \delta L_m \tag{A.10}$$

Note that δQ is just the angular momentum in the incoherent spin modes. Equation (A.9) for δS is the essential equation underlying considerations of reversibility and irreversibility and the second law. *The existence of a "heat" δQ need have nothing to do with energy*, although it almost always does.

REFERENCES

1. J. B. Boyling, *Comm. Math. Phys.* **10**, 52 (1968).
2. P. T. Landsberg, *Pure Appl. Chem.* **22**, 215 (1970).
3. L. Tisza, *Generalized Thermodynamics* (MIT Press, Cambridge, Massachussets, 1966).
4. H. Callen, *Thermodynamics* (Wiley, New York, 1960).
5. C. Truesdale, *Rational Thermodynamics* (McGraw-Hill, New York, 1969).

6. E. L. Hill, *Rev. Mod. Phys.* **23**, 253 (1951).
7. C. Palmieri and B. Vitale, *Nuovo Cimento A* **66**, 299 (1970).
8. Joe Rosen, *Ann. Phys.* **69**, 349 (1972).
9. E. P. Wigner, *Symmetries and Reflections* (Indiana Univ. Press, Bloomington, Indiana, 1967).
10. A. Aharony and Y. Ne'eman, *Int. J. Theoret. Phys.* **3**, 437 (1970).
11. A. A. Grib, E. V. Damaskinskii, and V. M. Maksimov, *Usp. Fiz. Nauk* **102**, 587 (1970) [English transl.: *Soviet Phys.—Uspekhi* **13**, 798 (1971)].
12. H. Wagner, *Z. Physik* **195**, 273 (1966).
13. P. W. Anderson, *Science* **177**, 393 (1972).
14. R. V. Lange, *Phys. Rev.* **146**, 301 (1966).
15. P. T. Landsberg and others, in *A Critical Review of Thermodynamics*, Ed. by E. B. Stuart, B. Gal-Or, and A. J. Brainard (Mono Book Corp., Baltimore, Maryland, 1970).
16. H. Callen and G. Horwitz, *Am. J. Phys.* **39**, 938 (1971).
17. R. B. Woodward and R. Hoffmann, *The Conservation of Orbital Symmetry* (Verlag Chemie, Wienheim and Academic Press, New York, 1970).
18. B. I. Halperin and P. C. Hohenberg, *Phys. Rev.* **188**, 898 (1969).
19. P. C. Martin, O. Parodi, and P. S. Pershan, *Phys. Rev. A* **6**, 2401 (1972).
20. W. Pauli, *Sommerfeld Festschrift*, Ed. by P. Debye (Hirzel, Leipzig, 1928).
21. N. G. van Kampen, *Physica* **20**, 603 (1954).
22. W. Heitler, *Ann. Inst. H. Poincaré* **15**, 67 (1956).
23. F. Coester, *Phys. Rev.* **84**, 1259 (1951).
24. R. H. Swendsen, Private communication.
25. B. J. Alder and T. E. Wainwright, *Phys. Rev. A* **1**, 18 (1970).
26. Ja. Sinai, *Statistical Mechanics Foundations and Applications, Proceedings of the I.U.P.A.P. Meeting, Copenhagen, 1966*, Ed. by T. A. Bak (Benjamin, New York, 1967); *Russian Math. Sur.* **25**, 137 (1970).
27. B. Gal-Or, *Science* **176**, 11 (1972).
28. K. G. Wilson, *Phys. Rev. B* **4**, 3174 (1971); **4**, 3184 (1971).
29. L. D. Landau, *Physik Z. Sowjetunion* **11**, 545 (1937).
30. L. D. Landau and E. M. Lifshitz, *Statistical Physics* (Pergamon Press, London, 1962).
31. E. Stanley, *Introduction to Phase Transitions and Critical Phenomena* (Oxford Univ. Press, New York and Oxford, 1971).
32. A. Michelson (Technion, Haifa), private communication.
33. C. Haas, *Phys. Rev.* **140A**, 863 (1965).
34. V. L. Ginzburg and L. P. Landau, *Zh. Eksperim. i Teor. Fiz.* **20**, 1064 (1950).
35. E. Schrödinger, *Statistical Thermodynamics* (Cambridge, 1936).
36. B. Widom, *J. Chem. Phys.* **43**, 3892, 3898 (1965).
37. H. Weyl, *The Theory of Groups and Quantum Mechanics*, Trans. by H. P. Robertson (Methuen, London, 1931 and Dover, New York, 1950).
38. R. B. Griffiths, in *A Critical Review of Thermodynamics*, Ed. by E. B. Stuart, B. Gal-Or, and A. J. Brainard (Mono Book Corp., Baltimore, Maryland, 1970).
39. H. Callen, in *Modern Developments in Thermodynamics*, Ed. by B. Gal-Or (Keter Publishing House, Jerusalem, 1973).
40. L. Onsager, *Phys. Rev.* **37**, 405 (1931); **38**, 2265 (1931).
41. H. Callen and T. Welton, *Phys. Rev.* **83**, 34 (1951).
42. R. Kubo, *J. Phys. Soc. Japan* **12**, 570 (1957).
43. K. M. Case, *Transport Theory and Statistical Physics* **2**(2), 129 (1972).
44. W. Bernard and H. Callen, *Rev. Mod. Phys.* **31**, 1017 (1959); *Phys. Rev.* **118**, 1466 (1960).